MIL-C-915/34B
23 August 1976
SUPERSEDING
MIL-C-915/34A
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE 7SGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

First - Copper, conductor, uncoated (See Details for size).
Second - Extruded silicone rubber insulation (See Details for diameter).
Third - Glass braid.
Fourth - Braid covering. Standard Identification Code, applied by method 1.
Fifth - The seven conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Non-fibrous fillers shall be used to form a firm, well-rounded assembly. Cabling sequence shall be consecutive, starting with No. 1 from center outward.
Sixth - Binder or combination binder/barrier.
Seventh - Special thermoplastic jacket (Gray) (See Details for thickness). Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) The type designation shall change to 7SGA.
(B) (b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(B) (e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage. When standard thermoplastic jacket is used, the air oven test does not apply.

Details

Type and size	Conductor size Navy Standard	Diameter over insulation nominal (nom) (inch)	Diameter over braid covering maximum (max) (inch)	Jacket thickness nom (inch)	Overall diameter max (inch)	Conductor resistance per 1000 ft at 25° Celsius (C) max (ohms)
7SGU-3	3(7)	0.096	0.130	0.040	0.545	4.0
7SGU-4	4(7)	.112	.143	.040	.595	2.51

(B) denotes changes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6145

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Voltage withstand - Volts, root mean square (Vrms), minimum (min)	3000
Insulation resistance - Megohms/1000 ft	500
Conductor resistance - Ohms/1000 ft, max	(See Details)
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Abrasion resistance - Scrapes, min	250
Crack resistance	No damage
Group B:	
Gas flame - 1 hour	No failure
Drip - At 95°C ± 1°C	Zero
Cold bending, cable - (Max mandrel diameter 12 times max cable diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - pound-force per square inch (lbf/in^2), min	750
Elongation - Percent, min	125
Jacket (Standard thermoplastic)	
Tensile strength - lbf/in^2, min	1800
Elongation - Percent, min	225
Jacket (special thermoplastic)	
(B) Tensile strength - lbf/in^2, min	2300
(B) Elongation - Percent, min	250
Group C:	
Physicals (aged)	
Jacket (standard thermoplastic)	
Hot oil immersion	
Tensile strength - lbf/in^2, min	1400
Elongation - Percent, min	50
Jacket (special thermoplastic)	
(B) Air oven (168 hours at 136°C)	
(B) Tensile strength - Percent of unaged, min	80
(B) Elongation - Percent of unaged, min	60
(B) Hot oil immersion	
(B) Tensile strength - Percent of unaged, min	75
(B) Elongation - Percent of unaged, min	50
Permanence of printing (conductor) - Cycles, min	50
Permanence of printing (jacket) - Cycles, min	250
Cable filler removability	

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of group A, B and C plus the following:

Cable aging and compatibility (125°C ± 5°C)	No failure
Cold working (minus 20°C ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, max	1.55

UNIT ORDERING LENGTHS:

All sizes 2000 feet (nom)

Custodians:
Army - MI
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH

(Project 6145-0674-34)

Review activities:
Army - EL, AV
Navy - EC

User activities:
Army - ME, MU, WC
Navy - CG

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